

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 02

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for the correct answer.</i> • $16.0 - 5.0 = \text{£}11 \text{ million}$ (1)	1
01.2	<i>1 mark for each point.</i> • It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided (1) • ALARA: emissions should be As Low As Reasonably Achievable (1) • BATNEEC: the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this (1)	3
01.3	<i>1 mark for each point.</i> • As industries expand or cities grow the number of pollutant sources increases (1) • So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed (1)	2

Question	Marking guidance	Marks
01.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Critical Pathway Analysis: predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk; mainly radioactive discharges; checked by sampling • Critical Group Monitoring: exposure of the most-exposed group monitored before health effects; if their risk is acceptable everyone else is safer; emissions reduced if not • Emission control strategies: location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) • Principles: polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship • Selection of methods: production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) • Evaluation: prevention usually cheaper and more effective than remediation; costs rise steeply near 100%; CPA relies on understanding of pathways; enforcement and international agreement needed for transboundary pollutants; judgement about acceptable risk is political as well as scientific • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

Question	Marking guidance	Marks
02.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming CIO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • Evidence of success: ozone hole area and depth stabilised and beginning to shrink; CFC concentrations in the atmosphere falling; UV levels not rising further; global ozone loss limited to about 4% • Reasons for success: international recognition of serious consequences (skin cancer); agreement of nearly every country; legally binding targets with a fund for developing countries; relatively few producers / products; alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap; no lifestyle change needed • Limits: persistence of CFCs means recovery takes decades; some ODSs still used; HFC replacements are greenhouse gases • Climate change contrast: greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country; fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations; emissions still rising; natural fluctuations, time delays and tipping points make cause-effect less clear to the public • Judgement: ozone control was a technical substitution problem; climate change is a systemic one, so the ozone success is only partly transferable	9
02.2	<i>1 mark.</i> • Vienna Convention for the Protection of the Ozone Layer - no legally binding goals (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	
03.1	<i>Correct answer only.</i> • B - Steam washing (hot water / steam jets kill organisms in the sediment) (1)	1
03.2	<i>1 mark for each point.</i> • Chemicals cause the oil molecules to join together / solidify (1) • Producing more solid material that can be collected / lifted more easily than liquid oil (1)	2

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> • Air contains about 210 cm³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm³ per litre (less than 3% of the amount in air) (1) • So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
04.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Raise river banks / build sea walls, flood barriers or wave screens (1) • Permeable urban surfaces (gravel, soil) instead of concrete / tarmac to slow runoff (1) • River flow management: upstream dams / low soil dams / afforestation / maintained flood plains to delay runoff (1) • Raised buildings on stilts or floating houses (1) • Pumping rainwater from land below river level (1) • Managed retreat - abandoning land where protection costs exceed value (eg Happisburgh) (1)	3
04.2	<i>1 mark for each point.</i> • Where the cost of flood prevention is greater than the value of what is protected, or defences would be ineffective (1) • Land is abandoned as water levels rise / coast erodes; defences used only where land is very important (1)	2
04.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Road surfaces deform as tar melts - need tarmac with a higher melting point (1) • Rail track expands and buckles above the temperature it was pre-stretched for (27 °C in UK) - relay with higher pre-stress (1) • Bridge damage from high river flows / debris; landslides on deforested slopes (1)	2
04.4	<i>1 mark for each point.</i> • CO₂ is captured from the flue gases (or carbon removed from the fuel before combustion) (1) • Transported by pipeline / ship / tanker and stored in depleted oil or gas fields, aquifers, or used for secondary oil recovery (1) • Limitation: developmental / expensive, uses energy (reducing efficiency), risk of leakage, limited storage sites (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 99-100, 105-108; AQA spec 3.2.1.2.	
05.1	<i>1 mark for each point.</i> • t-test (or Mann-Whitney U if not normally distributed) (1) • Comparing two means of counted / measured data from two independent samples (1)	2
05.2	<i>1 mark for each point.</i> • The difference is statistically significant (less than 1% probability that it arose by chance) (1) • The null hypothesis (no difference between fields) is rejected; the organic field has significantly more earthworms (1)	2

Question	Marking guidance	Marks
05.3	1 mark for each point. - A significant difference does not show a causal relationship (1) - Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference (1)	2
05.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Scientific methodology: identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research - Sample location: random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias; interval chosen so variation is detected without excess data - Sample timing: diurnal, seasonal, weather-related and long-term changes; sampling on different occasions; interval matched to rate of change - Sample size and number: larger and more samples for non-homogeneous variables; running mean and standard deviation show when the mean is reliable; preliminary study to decide - Standardisation and control of variables; accuracy, precision, reliability, representativeness; anomalies and repeats; calibration of equipment; appropriate techniques and risk management - Statistical analysis: choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation - Evaluation: trade-offs between cost, time, disturbance and reliability; limitations of sub-sampling; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	
06.1	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. - No reservoir range = $3.9 - 0.7 = 3.2$ m; with reservoir = $2.7 - 1.7 = 1.0$ m (1) - Reduction = 2.2 m (1)	2
06.2	1 mark for each point. - After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream (1) - During dry periods the dam is opened to release water and maintain river levels (1)	2

Question	Marking guidance	Marks
06.3	1 mark for each point. - Deforestation / urbanisation reduce infiltration and interception (1) - So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods (1)	2
06.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries - Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow - Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering - Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control - Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows - Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	
07.1	Correct answer only. Aerobic decomposition produces little methane and returns nutrients to the soil (1)	1

Question	Marking guidance	Marks
07.2	<i>1 mark for each point.</i> - It may contain concentrated heavy metals / toxic residues (1) - It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble (1)	2
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Impermeable polymer / clay liner to stop leachate escaping (1) - Collection and treatment of leachate (1) - Collection of methane and use as fuel (1) - Daily covering of waste / final capping and restoration (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
08.1	<i>Correct answer only.</i> B - Tropospheric ozone (formed by reactions of NO_x in sunlight) (1)	1
08.2	<i>1 mark for each point.</i> - Primary pollutant: released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores; NO_x from high-temperature combustion (1) - Secondary pollutant: produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) (1) - Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid; NO_x dissolve to form nitric acid (1) - The acids (sulfurous, sulfuric, nitric) are secondary pollutants; ozone itself is a secondary pollutant of NO_x photochemistry (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
09.1	<i>Correct answer only.</i> Oxygen (1)	1
09.2	<i>1 mark for each point.</i> - A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) (1) - There is still a lot that is not understood about how atmospheric processes work (1)	2
09.3	<i>1 mark for each point.</i> - The surface warmed in the tropics heats the air above; warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes (1) - Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: Lasky's principle - vast quantities exist at low grade; reserves vs resource; COOG affected by price, technology, energy costs • Improved exploration: higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • Low-energy extraction of low-grade ores: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • Previously inaccessible deposits: polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • Extending use: recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • Evaluation: environmental impacts vs benefits; energy costs; economics; which strategies are most sustainable; conclusion	9
10.2	<i>1 mark.</i> • All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	

Question	Marking guidance	Marks
11.1	Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Nature of the problem: interconnected atmosphere-hydrosphere-biosphere systems; changes over different time and spatial scales; natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change; time delay between cause and effect (ocean heat capacity) • Limitations of historic data: sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • Proxy data and their reliability: dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • Modern monitoring: long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty; ecological uncertainty (species survival, distribution, fragmentation) • Computer models: how they are tested and refined; importance of accurate representative data; scenario-based predictions and ranges • Uncertainty over ENSO / Conveyor / jet stream responses • Evaluation and conclusion: confidence in the overall trend and human cause despite detailed uncertainty; role of peer review and international bodies (IPCC) Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	25

Question	Marking guidance	Marks
11.2	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25